# Vectors I: Homework 3

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Vector product: properties, parallel vectors, areas, collinearity via cross product. Q1 True/False from the Basics Worksheet (Vector Product). Q2–Q7 are full promotional-examination questions. Total: 46 marks.

## Questions

### 1

For the following statements, state whether it is **true** or **false**.

| | **Statement** | **T/F** |
|---|---|---|
| **(a)** | $\mathbf{a}\times\mathbf{a}=\mathbf{0}$. | |
| **(b)** | $\mathbf{a}\times\mathbf{a}=0$. | |
| **(c)** | $\mathbf{i}\times\mathbf{j}=\mathbf{k}$. | |
| **(d)** | $\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$. | |
| **(e)** | $\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$. | |
| **(f)** | $\mathbf{a}\times\mathbf{b}$ is a scalar. | |
| **(g)** | $\mathbf{k}\times\mathbf{j}=\mathbf{i}$. | |
| **(h)** | $\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$. | |
| **(i)** | $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$. | |
| **(j)** | $(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$. | |
| **(k)** | For non-zero and non-parallel $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. | |
| **(l)** | $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar. | |
| **(m)** | $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where $\theta$ is the angle between $\mathbf{a}$ and $\mathbf{b}$. | |
| **(n)** | $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. | |
| **(o)** | If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. | |
| **(p)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$. | |
| **(q)** | The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $\lvert\mathbf{a}\times\mathbf{b}\rvert$. | |
| **(r)** | The area of triangle $OAB$ is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$. | |
| **(s)** | $\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector. | |
| **(t)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$. | |
| **(u)** | $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$. | |
| **(v)** | $\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$. | |
| **(w)** | $\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. | |
| **(x)** | $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$. | |
| **(y)** | The shortest distance from $B$ to the line through $A$ parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$. | |
| **(z)** | If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. | |

**2** [JPJC 2024 Promo Q11]

Referred to the origin $O$, points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. Point $C$ is the midpoint of $OA$ and point $D$ lies on $BC$ such that $BD:DC=2:3$.

**(i)** Find the position vectors $\overrightarrow{OC}$ and $\overrightarrow{OD}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. **[2]**

**(ii)** Find the area of triangle $OCD$ in the form of $k|\mathbf{a}\times\mathbf{b}|$, where $k$ is a constant to be found. **[3]**

**(iii)** Given that $|\mathbf{a}|=2$, $|\mathbf{b}|=\sqrt{2}$ and angle $AOB$ is $\frac{\pi}{4}$ radian, using scalar product, show that $CD$ is perpendicular to $OA$. **[3]**

**(iv)** Given instead that $\mathbf{a}$ is a unit vector, give a geometrical interpretation of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

**3** [MI 2024 Promo Q4]

Referred to the origin $O$, the points $A$, $B$ and $C$ have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. $C$ lies on $AB$ such that $AC:AB=1:3$.

**(i)** Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. **[2]**

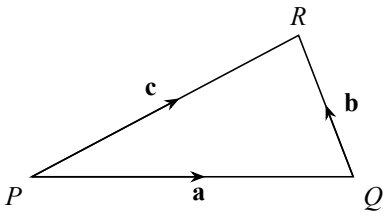
**(ii)** Show that the area of triangle $OAC$ is given by $k|\mathbf{a}\times\mathbf{b}|$, where $k$ is a constant to be determined. **[2]**

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

**(iii)** Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. **[3]**

**4** [ASRJC 2024 Promo Q5]

In triangle $PQR$, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ represent the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{PR}$ respectively.




**(a)** State a geometrical interpretation of $|\mathbf{c}\times\hat{\mathbf{a}}|$ and hence show that the area of the triangle $PQR$ can be written as $\frac{1}{2}|\mathbf{c}\times\mathbf{a}|$ units$^2$. **[3]**

**(b)** With clear explanations, show that $|\mathbf{c}\times\mathbf{a}|=|\mathbf{b}\times\mathbf{a}|$ and deduce that

$$\frac{|\mathbf{c}|}{\sin\angle PQR}=\frac{|\mathbf{b}|}{\sin\angle QPR} \quad \text{(Sine Rule)}.$$

**[2]**

**5** [CJC 2024 Promo Q7]

**(a)** Three non-zero vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are such that $3\mathbf{p}\times\mathbf{q}=\mathbf{r}\times\mathbf{p}$.
Find a linear relationship between $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$. **[3]**

**(b)** Referred to the origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$.
It is given that $\mathbf{a}$ and $\mathbf{b}$ are non-parallel unit vectors.
The point $B$ divides $AC$ in the ratio $1:3$.
The point $D$ lies on $OB$ produced such that $OB:OD=1:m$, where $m\in\mathbb{Z}$, $m\geq 2$.
Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, find the numerical value of $m$. **[4]**

**6** [SAJC 2024 Promo Q9]

**(a)** Relative to the origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

**(i)** By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. **[3]**

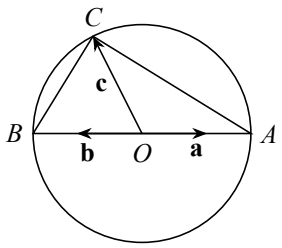
**(ii)** Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

**(iii)** The points $P$, $Q$ and $R$ have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of $\lambda$. **[3]**

**(b)** In the triangle $OUV$ where $O$ is the origin, the position vectors of the points $U$ and $V$ are $\mathbf{u}$ and $\mathbf{v}$ respectively. The point $W$ is the midpoint of $OU$, and the point $X$ has position vector given by $\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}$. Show that the area of triangle $UWX$ can be written as $m|\mathbf{u}\times\mathbf{v}|$ where $m$ is a constant to be found. **[3]**

**7** [CJC 2025 Promo Q6]

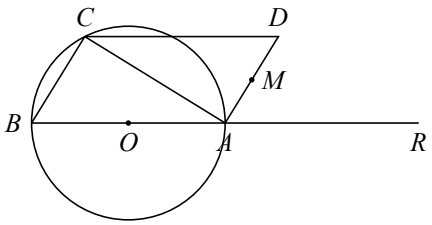
The diagram shows a unit circle with centre $O$ and diameter $AB$. The point $C$ lies on the circumference of the circle. The position vectors of $A$, $B$ and $C$ are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.




**(a)** By considering an appropriate scalar product, show that $AC$ and $BC$ are perpendicular. **[4]**

**(b)** Give a geometrical interpretation of $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$. **[1]**

The point $D$ is such that $ABCD$ is a parallelogram. The point $M$ is the mid-point of $AD$ and the point $R$ lies on $OA$ produced such that $OA:AR=1:2$.




**(c)** By considering $\overrightarrow{CM}\times\overrightarrow{CR}$ or otherwise, show that the points $C$, $M$ and $R$ are collinear. **[3]**

# Answers

**1** **(a)** T **(b)** F **(c)** T **(d)** F **(e)** T **(f)** F **(g)** F **(h)** T **(i)** T **(j)** T **(k)** T **(l)** T **(m)** T **(n)** F **(o)** F **(p)** T **(q)** T **(r)** F **(s)** F **(t)** T **(u)** T **(v)** T **(w)** F **(x)** F **(y)** F **(z)** T

**2** **(i)** $\overrightarrow{OC}=\frac{1}{2}\mathbf{a}$, $\overrightarrow{OD}=\frac{1}{5}(\mathbf{a}+3\mathbf{b})$

**(ii)** $\frac{3}{20}|\mathbf{a}\times\mathbf{b}|$, $k=\frac{3}{20}$

**(iii)** Shown.

**(iv)** $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$ (or a line or vector parallel to $\overrightarrow{OA}$).

**3** **(i)** $\mathbf{c}=\frac{2\mathbf{a}+\mathbf{b}}{3}$

**(ii)** Shown. $k=\frac{1}{6}$

**(iii)** 1

**4** **(a)** $|\mathbf{c}\times\hat{\mathbf{a}}|$ represents the perpendicular distance from the point $R$ to the line $PQ$. Shown.

**(b)** Shown.

**5** **(a)** $\mathbf{p}=k(3\mathbf{q}+\mathbf{r})$, $k\in\mathbb{R}\setminus\{0\}$

**(b)** $m=7$

**6** **(a)(i)** Shown.

**(a)(ii)** $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$.

**(a)(iii)** $\lambda=-25$

**(b)** $m=\frac{3}{16}$

**7** **(a)** Shown.

**(b)** $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$ is the length of projection of $\overrightarrow{AC}$ onto $\overrightarrow{OA}$.

**(c)** Shown.

**END OF WORKSHEET**